Sports Data Pipeline on AWS

Live and historical sports data · ingest → raw lake → clean lake → warehouse → models → dashboards

Architecture v1.1.0 · 12 Nov 2026 · see docs/architecture/CHANGELOG.md

Planned, not deployed yet

Live / streaming flow

Batch flow

Orchestration trigger / quality check

CI/CD deployment flow

ORCHESTRATION

EventBridge Scheduler
polls on a schedule

schedule

Amazon MWAA (managed Airflow)
or Step Functions for lower cost

DAGs trigger the batch backfill, Glue jobs, then dbt build + tests

SOURCES

Live REST APIs
API-Football · NHL API · MLB StatsAPI · balldontlie · OpenF1

Live feeds
SportScore websockets · OpenF1 live · OpenF1 history (replay)

Historical
StatsBomb Open Data · Jolpica / FastF1 · Cricsheet · nflverse · Kaggle

Reference API
TheSportsDB: teams, players (no live data on free plan)

INGESTION

AWS Lambda pollers
call the REST APIs, free tier ~100 calls/day

ECS Fargate task
websocket listeners + OpenF1 replay simulator

Batch backfill
Lambda or Glue job loads StatsBomb, Jolpica, Cricsheet and reference data into S3 raw

STREAMING

Kinesis Data Streams
live event stream

Secrets Manager
API keys for Lambda, Fargate and backfill

DELIVERY

Kinesis Data Firehose
buffers and writes JSON to S3

OpenF1 note
Live data only during sessions; none between sessions or in the off-season (Dec–Mar). Historical (2023+) is free; real-time may need a paid plan. The replay simulator feeds past sessions in time order to mimic a live stream.

RAW STORAGE

Amazon S3 · raw zone
JSON, partitioned by source / date

PROCESSING

AWS Lambda
Active: stand-in while Glue is blocked
sports-<env>-raw-to-clean
Python 3.13 · arm64 · AWS SDK for pandas layer (pyarrow)
JSON → Parquet at <table>/dt=<date>/
registers tables + partitions in the Catalog
on demand now; orchestrator starts it in phase 9

`processing_engine = "lambda" | "glue"`

AWS Glue job
Planned: waiting on AWS Support
Glue 5.0 · Spark · FLEX · 2× G.1X · same inputs and outputs
Job creation blocked ("Account is denied access"), case open
Deployed: IAM role, log group, code in artifacts bucket.
Missing: the job

CLEAN ZONE

Amazon S3 · clean zone
Parquet tables

Glue Data Catalog
table definitions

schemas

WAREHOUSE

Amazon Athena
low cost · or Redshift Serverless

MODELLING

dbt models
dbt-athena or dbt-redshift
fact_events · dim_team · dim_player · dim_session

OUTPUTS

Amazon QuickSight
dashboards

Notebooks
analysis

DATA QUALITY + MONITORING

Amazon CloudWatch
logs, metrics and alarms for Lambda, Fargate, Kinesis, Firehose, Glue and MWAA

Glue Data Quality
rules on the clean zone

Late or corrected events
keep the latest version per event id; reprocess the affected date partitions

dbt tests
not_null · unique · relationships · freshness

DEPLOYMENT · TERRAFORM + GITHUB CI/CD

GitHub repo
/infra · /ingestion · /dbt · /orchestration · /.github/workflows

GitHub Actions
CI/CD workflows

On pull request
terraform fmt + validate · Checkov / tfsec · plan posted to the PR · pytest for Lambdas · dbt build on dev

On merge to main
Docker image → ECR · package Lambdas · terraform apply to dev · sync DAGs to S3 · run dbt

OIDC → IAM role
assumed by Actions · no stored AWS keys

Terraform
modules in /infra

Terraform state
S3 backend with state locking · separate state for dev and prod

apply

AWS · dev
full pipeline stack

promote

Manual approval

Promote to prod
GitHub Environments approval, then terraform apply to prod

approved

AWS · prod
full pipeline stack